COMPUTER NETWORKS LAB

Packet Tracer Practical Guide
Experiments 2–9

Beginner-friendly procedures, addressing tables, router and switch commands, GUI configuration, verification and troubleshooting.

Experiment	Topic
2	Subnetting
3	LAN topology with DNS and DHCP
4	Static Routing
5	Dynamic Routing using OSPF
6	VLAN and Inter-VLAN Routing
7	IPv6 Autoconfiguration using SLAAC
8	Static NAT, Dynamic NAT and PAT
9	Bus, Star, Ring and Mesh topologies

Important: Interface names differ between Packet Tracer router models. Always check **show ip interface brief** and use the interfaces actually present on your device.

How to use this guide

• Build the topology first, then connect the cables, configure IP addresses, configure routers/switches, and test connectivity.

• Use Copper Straight-Through for PC-to-switch and usually router-to-switch/server Ethernet connections. Use Serial DCE for serial WAN links.

• Wait for links to initialize. Green indicators and an interface state of up/up normally indicate an operational link.

• Save configurations with **write memory** or **copy running-config startup-config**.

• Do not proceed to a higher-level feature such as NAT until the underlying interfaces, IP addressing and routing work.

Experiment 2 – Subnetting

Aim: Divide 192.168.10.0/24 into two /25 subnets and connect them through a router.

Devices and topology

• 1 Router (2911), 2 switches (2960), 6 PCs.

• PC0–PC2 → Switch0 → Router Fa0/0.

• PC3–PC5 → Switch1 → Router Fa1/0.

• Use Copper Straight-Through cables.

Subnet detail	Subnet 1	Subnet 2
Network	192.168.10.0	192.168.10.128
Mask	255.255.255.128 (/25)	255.255.255.128 (/25)
Usable host range	192.168.10.1–126	192.168.10.129–254
Broadcast	192.168.10.127	192.168.10.255
Usable hosts	126	126

Device	IP address	Gateway
Router Fa0/0	192.168.10.1	—
Router Fa1/0	192.168.10.129	—
PC0	192.168.10.2	192.168.10.1
PC1	192.168.10.3	192.168.10.1
PC2	192.168.10.126	192.168.10.1
PC3	192.168.10.130	192.168.10.129
PC4	192.168.10.131	192.168.10.129
PC5	192.168.10.254	192.168.10.129

Set every PC's subnet mask to **255.255.255.128** using Desktop → IP Configuration → Static.

Router configuration

```
enable
configure terminal
interface fastethernet 0/0
 ip address 192.168.10.1 255.255.255.128
 no shutdown
exit
interface fastethernet 1/0
 ip address 192.168.10.129 255.255.255.128
```

```
 no shutdown
exit
end
write memory
```

Verification

```
show ip interface brief
show ip route
```

From PC0, test **ping 192.168.10.1** and **ping 192.168.10.130**. Successful replies confirm local gateway and inter-subnet connectivity.

Experiment 3 – LAN with DNS and DHCP

Aim: Configure two LANs, a DNS server, a web server and a DHCP server.

Devices and topology

• 1 Router, 2 switches, DNS Server, Web Server, DHCP Server, PC clients and a laptop.

• LAN 1 (192.168.10.0/24): DNS, Web Server, PC2 and PC3 through Switch0 to Router Fa0/0.

• LAN 2 (192.168.20.0/24): DHCP Server and Laptop0 through Switch1 to Router Fa1/0.

Device	IP address	Gateway
Router Fa0/0	192.168.10.1	—
Router Fa1/0	192.168.20.1	—
DNS Server	192.168.10.2	192.168.10.1
PC2	192.168.10.3	192.168.10.1
Web Server	192.168.10.4	192.168.10.1
PC3	192.168.10.5	192.168.10.1
DHCP Server	192.168.20.2	192.168.20.1
Laptop0	DHCP	Automatic

Use subnet mask **255.255.255.0** for all static IPv4 devices.

Router configuration

```
enable
configure terminal
interface fastethernet 0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
interface fastethernet 1/0
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
end
write memory
```

DNS server

• Server → Desktop → IP Configuration: 192.168.10.2 / 255.255.255.0, gateway 192.168.10.1.

• Server → Services → DNS: switch DNS ON.

• Add an A record: Name www.myweb.com, Address 192.168.10.4.

Web server

• Set static IP 192.168.10.4, mask 255.255.255.0, gateway 192.168.10.1.

• Services → HTTP → ON.

DHCP server

• Set static IP 192.168.20.2, mask 255.255.255.0, gateway 192.168.20.1, DNS 192.168.10.2.

• Services → DHCP → ON. Create pool Net2 with gateway 192.168.20.1, DNS 192.168.10.2, start IP 192.168.20.10, mask 255.255.255.0, maximum users 50. Click Add/Save.

> Laptop0 → Desktop → IP Configuration → DHCP. It should receive an address from the DHCP pool.

Verification

```
ping 192.168.10.4
ping 192.168.10.2
ping www.myweb.com
```

On a client, open Desktop → Web Browser and visit **http://www.myweb.com**.

Experiment 4 – Static Routing

Aim: Connect two different LANs using two routers and static routes.

• R1 Fa0/0 → Switch1 → PC0 and PC1.

• R2 Fa0/0 → Switch2 → PC2 and PC3.

• R1 Serial2/0 ↔ R2 Serial2/0 using Serial DCE cable.

Device	IP address	Gateway
R1 Fa0/0	192.168.10.1/24	—
R1 Serial2/0	10.10.10.1/8	—
R2 Fa0/0	192.168.20.1/24	—
R2 Serial2/0	10.10.10.2/8	—
PC0	192.168.10.2	192.168.10.1
PC1	192.168.10.3	192.168.10.1
PC2	192.168.20.4	192.168.20.1
PC3	192.168.20.2	192.168.20.1

> The WAN mask /8 is what the supplied lab uses. /30 is more typical for a point-to-point link, but do not change the lab addressing unless instructed.

R1 configuration

```
enable
configure terminal
hostname R1
interface fastethernet 0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
interface serial 2/0
 ip address 10.10.10.1 255.0.0.0
 clock rate 64000
 no shutdown
exit
ip route 192.168.20.0 255.255.255.0 serial 2/0
end
write memory
```

Enter clock rate only on the DCE end. Check with **show controllers serial 2/0**.

R2 configuration

```
enable
configure terminal
hostname R2
interface fastethernet 0/0
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
interface serial 2/0
 ip address 10.10.10.2 255.0.0.0
 no shutdown
exit
ip route 192.168.10.0 255.255.255.0 10.10.10.1
end
write memory
```

Verification

```
show ip interface brief
show ip route
```

From PC0: **ping 192.168.20.2**. From PC3: **ping 192.168.10.2**. Both routers must have a route to the opposite LAN. Static routes appear with S in the routing table.

Experiment 5 – Dynamic Routing using OSPF

Aim: Allow routers to learn remote networks dynamically using OSPF. The following is a clear three-router example; confirm the addressing against your instructor's exact Experiment 5 diagram before using it.

• Three routers connected in a triangle; one PC/LAN behind each router.

• Use each router's actual Ethernet and serial interface names. The sample uses Fa0/0, Serial2/0 and Serial3/0.

• PC gateways: 192.168.1.1, 192.168.2.1 and 192.168.3.1 respectively.

Link	Address at first end	Address at second end
LAN 1	R0 Fa0/0: 192.168.1.1/24	PC: 192.168.1.2/24
LAN 2	R1 Fa0/0: 192.168.2.1/24	PC: 192.168.2.2/24
LAN 3	R2 Fa0/0: 192.168.3.1/24	PC: 192.168.3.2/24
R0–R1	10.0.0.1/30	10.0.0.2/30
R0–R2	10.0.0.5/30	10.0.0.6/30
R1–R2	10.0.0.9/30	10.0.0.10/30

Configure router interfaces

Router0:

```
enable
configure terminal
interface fastethernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
exit
interface serial 2/0
 ip address 10.0.0.1 255.255.255.252
 clock rate 64000
 no shutdown
exit
interface serial 3/0
 ip address 10.0.0.5 255.255.255.252
 clock rate 64000
 no shutdown
exit
end
```

Router1:

```
enable
```

```
configure terminal
interface fastethernet 0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
exit
interface serial 2/0
 ip address 10.0.0.2 255.255.255.252
 no shutdown
exit
interface serial 3/0
 ip address 10.0.0.9 255.255.255.252
 clock rate 64000
 no shutdown
exit
end
```

Router2:

```
enable
configure terminal
interface fastethernet 0/0
 ip address 192.168.3.1 255.255.255.0
 no shutdown
exit
interface serial 2/0
 ip address 10.0.0.6 255.255.255.252
 no shutdown
exit
interface serial 3/0
 ip address 10.0.0.10 255.255.255.252
 no shutdown
exit
end
```

Clock rate is needed only on the DCE end of each serial link. The interface mapping above is an example; check the actual cable endpoints.

Configure OSPF process 2

Router0:

```
configure terminal
router ospf 2
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0
end
write memory
```

Router1:

```
configure terminal
router ospf 2
 router-id 2.2.2.2
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
end
write memory
```

Router2:

```
configure terminal
router ospf 2
 router-id 3.3.3.3
 network 192.168.3.0 0.0.0.255 area 0
 network 10.0.0.4 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
end
write memory
```

Verification

```
show ip interface brief
show ip route
```

```
show ip protocols
```

Look for O routes in the routing table. From PC1, ping 192.168.3.2. If no O routes appear, check interfaces, serial links, network statements and OSPF neighbor status.

Experiment 6 – VLAN and Inter-VLAN Routing

Aim: Separate student and faculty computers into VLANs and route between them using router-on-a-stick.

• 1 router, 1 2960 switch, 4 PCs.

• PC-A → Fa0/1; PC-B → Fa0/2; PC-C → Fa0/3; PC-D → Fa0/4.

• Switch Fa0/24 → Router Gi0/0.

PC	VLAN	IP address	Gateway
PC-A	10	192.168.1.2/24	192.168.1.1
PC-B	10	192.168.1.3/24	192.168.1.1
PC-C	20	172.16.1.2/24	172.16.1.1
PC-D	20	172.16.1.3/24	172.16.1.1

Switch VLAN configuration

```
enable
configure terminal
vlan 10
 name STUDENT
exit
vlan 20
 name FACULTY
exit
interface range fastethernet 0/1 - 2
 switchport mode access
 switchport access vlan 10
exit
interface range fastethernet 0/3 - 4
 switchport mode access
 switchport access vlan 20
exit
interface fastethernet 0/24
 switchport mode trunk
 switchport trunk allowed vlan 10,20
end
write memory
```

Router subinterfaces

```
enable
configure terminal
interface gigabitethernet 0/0
 no ip address
 no shutdown
exit
interface gigabitethernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.1 255.255.255.0
exit
interface gigabitethernet 0/0.20
 encapsulation dot1Q 20
 ip address 172.16.1.1 255.255.255.0
exit
end
write memory
```

If your router uses a different physical port, replace Gi0/0 and the subinterface names consistently.

Verification

```
show vlan brief
```

```
show interfaces trunk
show ip interface brief
```

From PC-A, ping 192.168.1.3 (same VLAN) and 172.16.1.2 (other VLAN).

Experiment 7 – IPv6 Autoconfiguration using SLAAC

Aim: Enable IPv6 routing advertisements so PCs automatically create IPv6 addresses.

The lab text lists two routers but its described layout shows two separate LANs and no router-to-router link. This guide demonstrates SLAAC on each separate LAN. Without a link and routing between routers, the two LANs cannot communicate with one another.

LAN	Router interface IPv6 prefix	Router address
LAN 1	2001:AA::/64	2001:AA::1/64
LAN 2	2002:BB::/64	2002:BB::1/64

Router0 (LAN 1)

```
enable
configure terminal
ipv6 unicast-routing
interface gigabitethernet 0/0
 ipv6 address 2001:AA::1/64
 no shutdown
exit
end
write memory
```

Router1 (LAN 2)

```
enable
configure terminal
ipv6 unicast-routing
interface gigabitethernet 0/1
 ipv6 address 2002:BB::1/64
 no shutdown
exit
end
write memory
```

Use the actual router interface connected to each switch. If Router1 connects using Gi0/0, configure Gi0/0 instead of Gi0/1.

Configure each PC

• PC → Desktop → IP Configuration → IPv6 Auto Config.

• LAN 1 PCs should receive global addresses beginning 2001:AA::.

• LAN 2 PCs should receive global addresses beginning 2002:BB::.

Verification

```
ipconfig
ping <other-PC-global-IPv6-address>
```

Test PC0 to PC1 within LAN 1, and PC2 to PC3 within LAN 2. A link-local default gateway beginning FE80:: is normal for SLAAC.

Experiment 8 – Static NAT, Dynamic NAT and PAT

Aim: Translate private LAN addresses to external addresses using three NAT methods. Build and verify the base network before configuring NAT.

• PC1 and PC2 → Switch → R1 Fa0/0.

• R1 Serial2/0 ↔ R2 Serial2/0 using Serial DCE cable.

• R2 Fa0/0 → Server0 Fa0 using Ethernet cable.

Device/interface	IP address	Mask / gateway
PC1	192.168.1.10	/24, GW 192.168.1.1
PC2	192.168.1.11	/24, GW 192.168.1.1
R1 Fa0/0	192.168.1.1	255.255.255.0
R1 Serial2/0	200.10.10.1	255.255.255.252
R2 Serial2/0	200.10.10.2	255.255.255.252
R2 Fa0/0	10.0.0.1	255.0.0.0
Server0	10.0.0.10	/8, GW 10.0.0.1

R1 configuration

```
enable
configure terminal
hostname R1
interface fastethernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
 ip nat inside
exit
interface serial 2/0
 ip address 200.10.10.1 255.255.255.252
 no shutdown
 ip nat outside
exit
ip route 0.0.0.0 0.0.0.0 200.10.10.2
end
write memory
```

If R1 is the DCE end, add **clock rate 64000** under Serial2/0.

R2 configuration

```
enable
configure terminal
hostname R2
interface serial 2/0
 ip address 200.10.10.2 255.255.255.252
 no shutdown
exit
interface fastethernet 0/0
 ip address 10.0.0.1 255.0.0.0
 no shutdown
exit
ip route 192.168.1.0 255.255.255.0 200.10.10.1
end
write memory
```

Server configuration

Server0 → Desktop → IP Configuration → Static: IP 10.0.0.10, mask 255.0.0.0, gateway 10.0.0.1.

Base connectivity check

```
show ip interface brief
show ip route
ping 200.10.10.2
ping 10.0.0.10
```

Do not configure NAT until R2's Ethernet interface to Server0 is up/up and the server responds to a ping from R2.

Static NAT example

```
configure terminal
```

```
ip nat inside source static 192.168.1.10 200.10.10.10
end
```

This maps inside-local 192.168.1.10 to inside-global 200.10.10.10. Caution: 200.10.10.10 is outside the configured 200.10.10.0/30 WAN subnet. Additional public-address routing or addressing is required for end-to-end reachability; the single mapping command alone is not enough.

Dynamic NAT example

```
configure terminal
no ip nat inside source static 192.168.1.10 200.10.10.10
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat pool NAT_POOL 203.0.113.5 203.0.113.10 netmask 255.255.255.248
ip nat inside source list 1 pool NAT_POOL
end
```

The sample pool 203.0.113.5–203.0.113.10 must be routed and available to R1. It is not automatically reachable through the /30 WAN.

PAT example

```
configure terminal
no ip nat inside source list 1 pool NAT_POOL
no ip nat pool NAT_POOL
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface serial 2/0 overload
end
write memory
```

PAT allows multiple private hosts to share the R1 Serial2/0 interface address. Test traffic from PC1 to the external server, then inspect translations.

NAT verification

```
show ip nat translations
show ip nat statistics
show access-lists
show ip route
```

Static NAT, Dynamic NAT and PAT are separate demonstrations. Remove the previous rule before testing another method. NAT translations generally appear only after matching traffic is generated.

Experiment 9 – Bus, Star, Ring and Mesh Topologies

Aim: Create common network layouts and test connectivity between end devices.

Common PC addressing

PC	IP address	Mask	Gateway
PC1	192.168.1.10	255.255.255.0	Blank
PC2	192.168.1.11	255.255.255.0	Blank
PC3	192.168.1.12	255.255.255.0	Blank
PC4	192.168.1.13	255.255.255.0	Blank

Bus and Star

In Packet Tracer, a simple switch-centered arrangement is often used to represent a logical bus or star in beginner labs. For a star, connect each PC directly to one central switch. For the bus demonstration, follow the specific diagram supplied by your instructor; Packet Tracer does not provide a traditional coaxial bus cable by default.

• Star: one central switch, four PCs; each PC connects directly to the switch using Copper Straight-Through.

• Bus: reproduce the instructor's specified shared-medium arrangement. If using a switch as a practical approximation, connect all PCs to that central switch and label the topology clearly.

Ring

• Place four switches and connect them in a closed loop: S1–S2–S3–S4–S1.

• Connect one PC to each switch.

• Use Copper Straight-Through for PC-to-switch and Copper Cross-Over or automatic cable selection for switch-to-switch links.

• Be aware that Layer 2 switches may block a redundant ring link through spanning tree; that is normal.

Mesh

• For a three-switch full mesh, connect S1 to S2, S2 to S3, and S1 to S3.

• Connect one PC to each switch.

• Use Copper Straight-Through for PC-to-switch and Copper Cross-Over or automatic cable selection for switch-to-switch links.

Verification

```
ping 192.168.1.11
ping 192.168.1.12
ping 192.168.1.13
```

On a switch, use **show mac address-table** and **show vlan brief**. PCs on the same IP subnet and VLAN should communicate if the links are operational.

Quick command reference

Task	Command
Enter privileged mode	enable
Global configuration	configure terminal
Enter interface	interface fa0/0
Assign IPv4 address	ip address IP MASK
Enable interface	no shutdown
Check interfaces	show ip interface brief
Check routing table	show ip route
Save configuration	write memory
Check VLANs	show vlan brief
Check trunk ports	show interfaces trunk
Check OSPF	show ip protocols
Check NAT translations	show ip nat translations
Check NAT statistics	show ip nat statistics
Test connectivity	ping IP_ADDRESS
Check serial DCE/DTE	show controllers serial 2/0

Common mistakes and fixes

• Forgot no shutdown: enter interface configuration and issue no shutdown.

• Wrong interface name: run show ip interface brief and use the exact interface shown.

• Duplicate IP address: each router interface and host must have a unique address in its subnet.

• Wrong PC gateway: set the gateway to the router interface in that PC's LAN.

• Missing reverse static route: both routers need routes to the remote LAN.

• Wrong OSPF wildcard: /24 uses 0.0.0.255; /30 uses 0.0.0.3.

• VLAN trunk missing: configure the switch port connected to the router as a trunk.

• SLAAC not working: ensure ipv6 unicast-routing is enabled and the router interface is up.

• Serial link down: check cable endpoints, IP/mask on both ends, no shutdown, and clock rate on DCE only.

• Ethernet shows up/down: check the connected device, cable endpoints, port status and device power. An empty server IP does not itself cause Ethernet line protocol to go down.

• NAT fails: first prove the interfaces and routes work without NAT; then configure one NAT method at a time.

Exam practice order

Suggested order: Experiment 9 (topologies), Experiment 2 (subnetting), Experiment 3 (DNS/DHCP), Experiment 4 (static routing), Experiment 6 (VLAN), Experiment 7 (SLAAC), Experiment 5 (OSPF), Experiment 8 (NAT).

This document is a study guide, not a replacement for checking your instructor's exact diagram and addressing table. Where a lab record is inconsistent or incomplete, the relevant caveat is stated in the experiment section.